

Reading and Plotting Points on a Grid

Answer Key

Grade 4–5

Quick Answers

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|---------|------|----------|-------|
| 1. 2, 5 | 4. 5 | 7. 6, 8 | 10. — |
| 2. 6, 3 | 5. 7 | 8. 10 | |
| 3. 0, 6 | 6. 5 | 9. 25, 5 | |
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Curriculum

Level: Grade 4–5

4.MD / 5.MD – *problem 9*

5.G.A.1–A.2 / 5.OA.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

6. *If they got -2: subtracted the two x -coordinates instead of the two y -coordinates*
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What this sheet is teaching, and what to listen for. There is one idea underneath all ten problems: an ordered pair is a pair of *instructions*, and the order of the two instructions is part of the meaning. Across first, then up. Nearly every error on a sheet like this is the same error — the two numbers used in the wrong order — and it is invisible in the answer $(7, 3)$ unless you watch the child’s finger. When a point lands in the wrong place, do not point at the grid; ask them to say the journey out loud.

1. Read point A .

Starting at the corner marked 0, trace across until you are underneath A : that is 2 squares. Now trace up: that is 5 squares.

Written as an ordered pair, $A = (2, 5)$.

Say it as a journey. “Start at zero, across two, up five.” A child who can say that sentence has the idea; a child who reads the two numbers off without moving a finger may be guessing.

2. Read point B .

Across 6, up 3, so $B = (6, 3)$.

The extra practice: $C = (4, 8)$ and $D = (9, 1)$.

B and C are the interesting pair. B is $(6, 3)$ and C is $(4, 8)$. B is further across but C is much higher. If a child writes $C = (8, 4)$ they have swapped the order — and $(8, 4)$ is a real point on this grid, which is exactly why the mistake survives: it does not look wrong, it looks like a different answer.

3. A point that sits on a line. E .

E sits right on the upright arrow, so you do not travel across at all before going up: across $\boxed{0}$, up $\boxed{6}$. So $E = (0, 6)$.

E is on the y -axis — the upright arrow.

Zero as an answer. Children often feel a coordinate of 0 must be a mistake, because “you didn’t do anything”. Zero is a real instruction: it says *do not go across*. Point out that the corner itself is $(0, 0)$ — both instructions are zero, and it is still a point.

4. Plot two points, then measure across. $P(3, 6)$ and $Q(8, 6)$.

P : across 3, up 6. Q : across 8, up 6. Both are six squares up, so $\overline{PQ}$ is a flat line.

Length: x of Q minus x of P , that is $\boxed{8} - \boxed{3} = \boxed{5}$ squares.

Why the y -coordinates were not needed. They are equal, so they cancel: the two points are at the same height and the only difference between them is how far across they are. Ask the child what would happen if Q were $(8, 7)$ instead — the line would slope, and counting across would no longer give its true length.

5. Now measure up. $R(4, 2)$ and $S(4, 9)$.

Both points are four squares across, so $\overline{RS}$ is upright.

Length: y of S minus y of R , that is $\boxed{9} - \boxed{2} = \boxed{7}$ squares.

The coordinate that is the same for both points is the x -coordinate, 4.

Pairing problems 4 and 5 on purpose. One flat line, one upright line, and in each case only one of the two coordinates does any work. Once a child can say *which* coordinate to subtract and why, they are not memorising two rules — they are reading the picture.

6. How much higher? $C(4, 8)$ and $B(6, 3)$.

“Higher” means going up, and going up is the *second* number in each pair.

y of C minus y of B : $8 - 3 = \boxed{5}$. So C is five squares higher than B .

The distractor is deliberate. B is further across than C (6 against 4), so a child working from the first numbers gets $4 - 6 = -2$, or writes 2 and reports that C is two squares higher. Neither is the answer to the question asked. Being further across has nothing to do with being higher — have them put one finger on each point and move both fingers straight down to the bottom arrow; the fingers stay the same distance apart vertically the whole way.

7. Two rules make points.

Rule A gives 0, 1, 2, 3, 4. Rule B adds 2 each time, giving 0, 2, 4, 6, 8 — so the missing table entries are 4, $\boxed{6}$ and $\boxed{8}$.

The five ordered pairs are $(0, 0)$, $(1, 2)$, $(2, 4)$, $(3, 6)$ and $(4, 8)$. Plotted and joined, they fall on one straight line rising to the right.

Every y is always **two** times the x beside it.

Where the relationship comes from. Rule B adds 2 every time Rule A adds 1, so y climbs twice as fast as x — and that is what makes the points line up. Ask the child to predict the pair for $x = 10$ before they check it.

8. Keep the pattern going.

The last pair was (4, 8). Rule A adds 1: the next x is 5. Rule B adds 2: the next y is 10. So the next point is (5, 10), and yes — it lands on the same straight line as the other five.

Two ways to get it, and both are right. A child can add on ($8 + 2 = 10$) or use the relationship ($2 \times 5 = 10$). The second way is the one that pays later, because it answers “what is y when $x = 50$?” without writing out fifty rows. If they used the adding-on method, ask them that question and let them discover the shortcut.

9. A graph that tells a story.

Reading the height of each point: Week 1 is 3, Week 2 is 5, Week 3 is 4, Week 4 is 8, Week 5 is 5.

Altogether: $3 + 5 + 4 + 8 + 5 =$ 25 books.

Shared equally over five weeks: $25 \div 5 =$ 5 books per week.

Maya’s best week was Week 4, with 8 books.

The mean is worth a sentence. Five books a week is the number Maya *would* have read each week if the reading had been spread out evenly — it is not a week that actually happened. Two of her weeks were exactly five, one was well above and two were below. Ask which she would rather tell someone about, and why both numbers are honest.

10. Explain it to a friend. (Open explanation — listen to the reason.)

(3, 7) means across 3 then up 7. (7, 3) means across 7 then up 3. They are two different journeys from the same corner, so they end in two different places — as the two marked points on the grid show.

A model answer. “They are not the same point. The first number always tells you how far to go across and the second tells you how far to go up, so if you swap them you swap the two journeys. For (3, 7) you go a little way across and a long way up, and you end up near the top on the left. For (7, 3) you go a long way across and only a little way up, and you end up near the bottom on the right. The numbers are the same but the *jobs* are different, which is why it is called an *ordered* pair.”

Marking guide. Full credit for any answer that says what the order *means* — first number across, second number up — or that describes the two different journeys. Partial credit for “they are in a different order so they are different points” with no account of what the order does. Not sufficient: “because the grid shows two dots” — true, but it repeats the picture instead of explaining it.

If the child insists they are the same: have them do both journeys with a finger, slowly, saying each step. The two fingers end up in obviously different places, and that usually settles it faster than any explanation.

Open explanation — open response.